SCIENCE DISCOVERY LAB

Basic 2 — Learner’s Guide

25 curriculum-aligned lessons • Observe • Explore • Explain • Practise

A colourful first science book about the natural world, materials, weather, water, energy, our bodies and safe everyday living.

Use this book with a teacher, parent or another trusted adult. All written explanations and questions are original educational material based on the uploaded NaCCA Basic 2 Science curriculum.

Your discovery journey

Diversity of Matter

1. Parts of a Green Plant • 2. Animal Body Parts • 3. Living or Non-living? • 4. Explore Material Properties • 5. Solids, Liquids and Gases • 6. Separating Solid Mixtures

Cycles

7. Wet and Dry Seasons • 8. Sources of Light • 9. Choosing Safe Drinking Water • 10. How We Use Air

Systems

11. What Our Body Parts Do • 12. Sunlight and Warmth

Forces and Energy

13. Energy in Everyday Life • 14. Gaining and Losing Heat • 15. Electrical Safety First • 16. Batteries Power Devices • 17. How Forces Change Movement • 18. Simple Machines at Work

Humans and the Environment

19. Keep Your Body Clean • 20. Caring for a Clean Classroom • 21. Ringworm and Healthy Skin • 22. Protecting Against Water-borne Diseases • 23. Technology in Our Community • 24. Preparing Food Safely • 25. Protect Our Environment

Safety comes first

Do not touch broken glass, unknown substances, electrical sockets or hot objects. Never taste materials during experiments. Water, food and hygiene activities need appropriate adult guidance. During storms, stay in safe shelter.

QUEST 1: Parts of a Green Plant

Diversity of Matter | Identify and explain the functions of roots, stem, leaves and flowers.

LET’S LEARN

A plant has several parts with different jobs. Roots usually hold it in the soil and take in water. The stem supports the plant and carries water to other parts. Green leaves use sunlight to make food. Many plants produce flowers, which help them make seeds. Plants may have different shapes but these four parts are useful to observe. Be gentle with growing plants.

Science words to remember

Roots • Stem • Leaves • Flowers

SCIENCE DISCOVERY • BASIC 2 • QUEST 01

Parts of a Green Plant

STEP 1 / 4



OBSERVE

A whole bean plant with roots, stem, leaves and flower is displayed.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Observe a bean plant or a large drawing. Label roots, stem, leaves and flowers. Draw arrows to show water travelling from the roots to the leaves.

Safety note: Do not uproot a community plant without teacher permission; use a drawing or a potted seedling.

QUEST 1: Show what you know

Observe the four teaching steps

1. OBSERVE: A whole bean plant with roots, stem, leaves and flower is displayed.

2. LABEL: Point to roots beneath the soil and stem above it.

3. TRACE: Show water moving from roots up the stem to leaves.

4. MATCH: Label leaves and flowers, then sketch the complete plant.

Guided practice — answer in your exercise book

1. Which plant part is usually under the soil?
2. Which part helps anchor a plant?
3. Which part holds many leaves upright?
4. Which part carries water through the plant?
5. What do green leaves use to make food?
6. Which part usually has a broad green surface?
7. Which plant part often helps produce seeds?
8. Where can you see flowers on many plants?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 2: Animal Body Parts

Diversity of Matter | Identify the head, trunk and limbs of familiar animals.

LET’S LEARN

Many familiar animals have a head, trunk and limbs. The head holds parts such as eyes, ears and mouth. The trunk is the main central body. Limbs are legs or arms that often help animals move or hold things. Goats, chickens and people do not have exactly the same shapes. Compare their bodies carefully rather than assuming all animals have four legs.

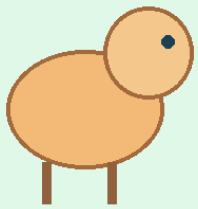
Science words to remember

Head • Trunk • Limbs • Compare

SCIENCE DISCOVERY • BASIC 2 • QUEST 02

Animal Body Parts

STEP 1 / 4



LOOK

Show a goat silhouette and a chicken silhouette side by side.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

Draw a goat and a chicken. Label head, trunk and limbs. Point out one difference between the two animals.

Safety note: Work with your teacher or a trusted adult.

QUEST 2: Show what you know

Observe the four teaching steps

1. LOOK: Show a goat silhouette and a chicken silhouette side by side.

2. HEAD: Highlight the eyes and mouth of each animal.

3. BODY: Outline each animal’s trunk.

4. LIMBS: Mark legs and wings and compare their use.

Guided practice — answer in your exercise book

1. Where are a goat’s eyes?
2. Which animal part often includes mouth and nose?
3. What is the central part of an animal body called?
4. Which part connects many limbs to the body?
5. What do a goat’s legs help it do?
6. How many legs does a healthy chicken usually have?
7. Do a goat and a chicken have identical body shapes?
8. Which animal would have wings rather than four walking legs?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 3: Living or Non-living?

Diversity of Matter | Sort things from the environment and explain their group.

LET’S LEARN

Living things carry out life processes such as growing and needing air or nourishment. Examples include grass, ants, goats and mango trees. Non-living things include stones, plastic containers and metal spoons. A toy car can move when pushed but it is not alive. A dry leaf was once part of a living tree, yet it is no longer living. Give a reason for every classification.

Science words to remember

Living • Non-living • Evidence • Sorting

SCIENCE DISCOVERY • BASIC 2 • QUEST 03

Living or Non-living?

STEP 1 / 4



COLLECT

Show a leaf, goat, stone, plastic cup and growing grass.

LOOK → NOTICE → EXPLAIN → TRY

Try it for yourself

On a supervised nature walk, record six items. Draw two columns labelled Living and Non-living and explain your choice to a classmate.

Safety note: Do not touch unknown insects, sharp waste or unsafe objects on a nature walk.

QUEST 3: Show what you know

Observe the four teaching steps

1. COLLECT: Show a leaf, goat, stone, plastic cup and growing grass.

2. SORT: Place goat and grass in the living column.

3. SORT: Place stone and plastic cup in the non-living column.

4. EXPLAIN: Movement alone does not prove that something is living.

Guided practice — answer in your exercise book

1. Which one is alive?
2. Which item needs water to keep growing?
3. Which is non-living?
4. Which is not alive even if it can roll?
5. Why is an ant living?
6. Why is a chair non-living?
7. Where does a living goat belong?
8. Where does a metal key belong?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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QUEST 4: Explore Material Properties

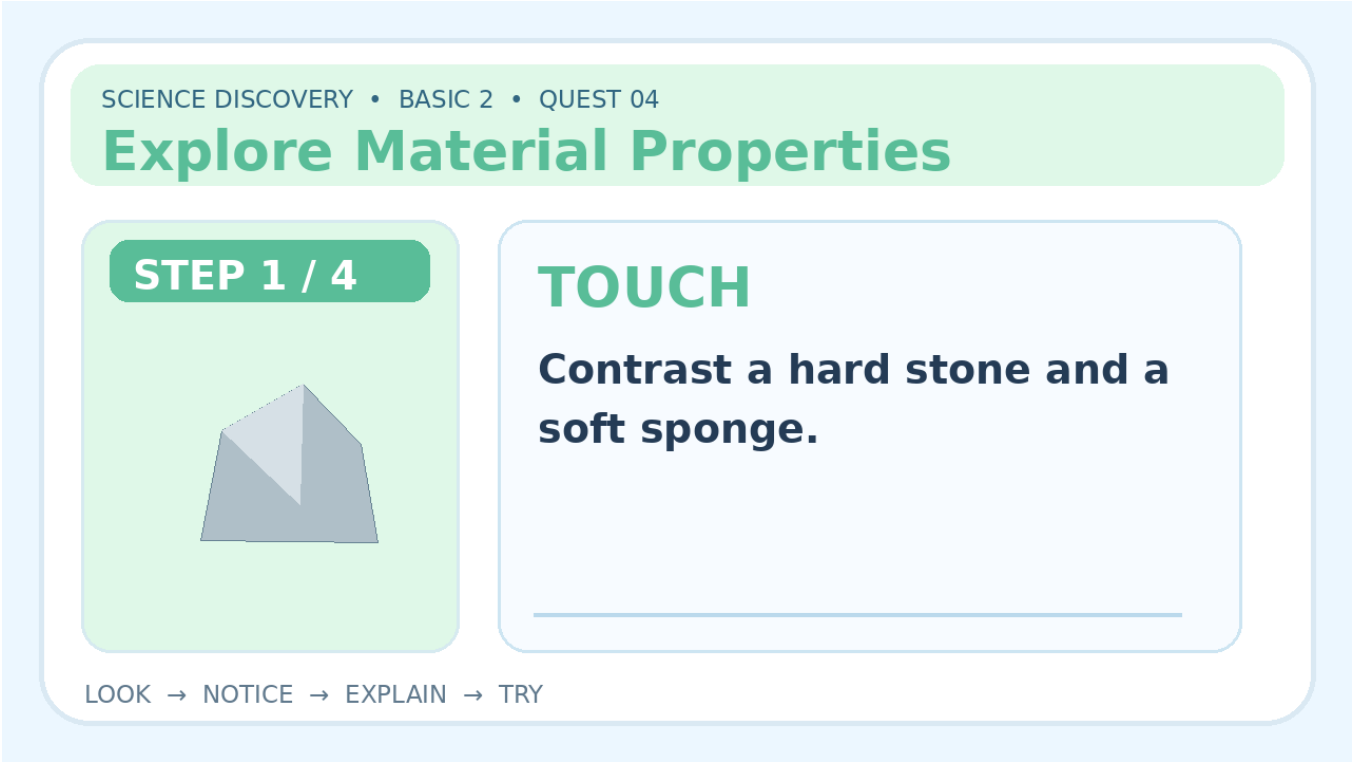
Diversity of Matter | Classify common materials as hard, soft, rough, smooth, opaque, transparent or bendable.

LET’S LEARN

Materials have observable properties. Cotton wool is soft; many stones are hard. Sandpaper can feel rough, while a polished tile can feel smooth. We can see through clear glass, so it is transparent; a wooden board is opaque because we cannot see through it. Some flexible plastic bends easily. One material can have more than one property. Choose safe objects to compare.

Science words to remember

Hard or soft • Rough or smooth • Transparent or opaque • Bendable



Try it for yourself

Safely compare paper, cotton wool, a wooden block and a clear plastic sheet. Draw a sorting table and put each in its suitable groups.

Safety note: Use unbroken plastic rather than sharp glass; do not bend or break unknown materials.

QUEST 4: Show what you know

Observe the four teaching steps

1. TOUCH: Contrast a hard stone and a soft sponge.

2. FEEL: Compare rough sandpaper with a smooth tile.

3. LOOK: Compare a clear sheet with an opaque board.

4. BEND: Show a flexible strip bending and returning to shape.

Guided practice — answer in your exercise book

1. Which material is usually soft?
2. Which item is normally hard?
3. Which surface usually feels rough?
4. Which surface can feel smooth?
5. Which material lets you see through clearly?
6. Which material blocks your view?
7. What can a flexible plastic strip do?
8. Which object bends without breaking easily?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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QUEST 5: Solids, Liquids and Gases

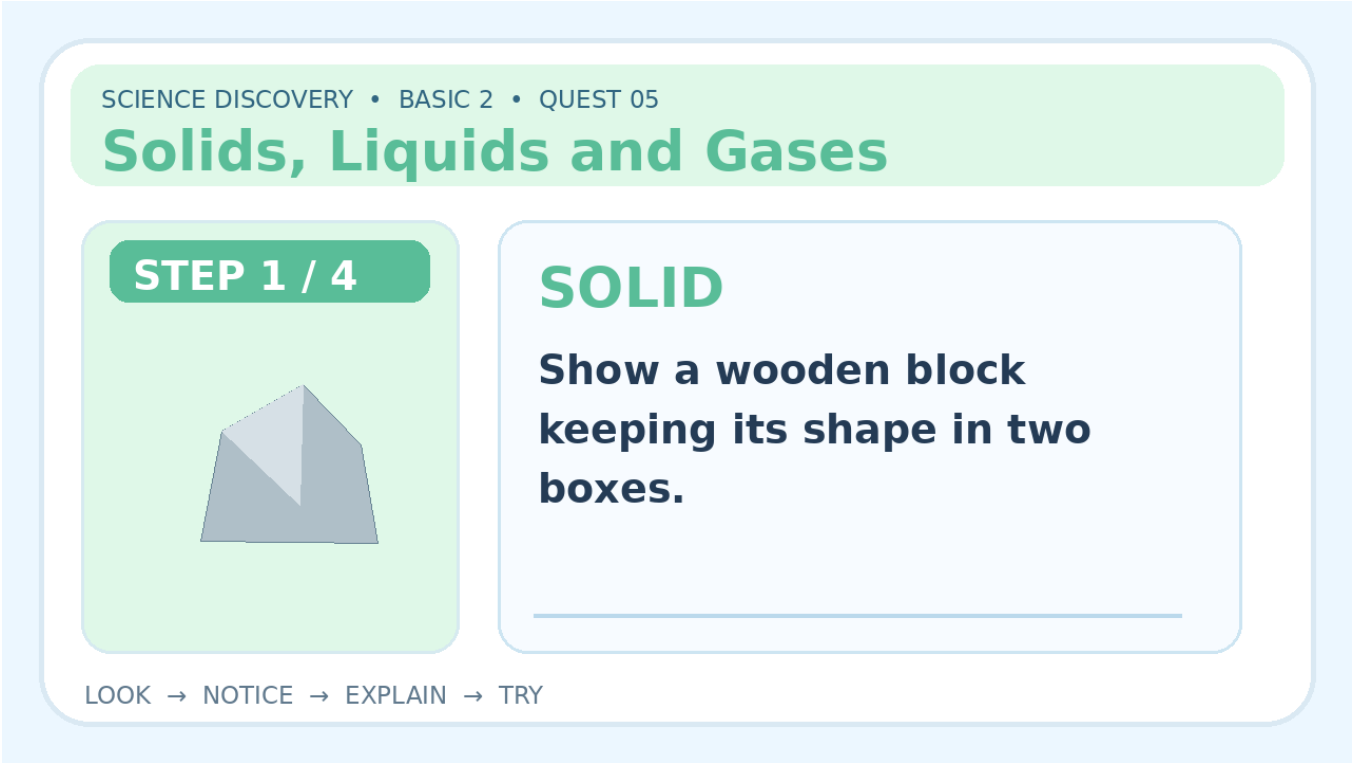
Diversity of Matter | Describe and compare how solids, liquids and gases behave.

LET’S LEARN

A solid usually keeps its own shape; a stone remains a stone when moved from a table to a bowl. A liquid flows and takes the shape of its container; water looks different in a bottle and a cup. A gas spreads to fill the space available; air inside a balloon has no fixed shape of its own. We can safely watch examples without tasting or smelling unknown substances.

Science words to remember

Solids • Liquids • Gases • Compare



Try it for yourself

Observe a block, water in a sealed clear container, and an inflated balloon. Record how each behaves and draw its shape.

Safety note: Do not taste liquid samples or use hazardous gases or materials.

QUEST 5: Show what you know

Observe the four teaching steps

1. SOLID: Show a wooden block keeping its shape in two boxes.

2. LIQUID: Move water from a wide bowl into a tall cup.

3. GAS: Show air filling a rounded balloon.

4. COMPARE: Label solid, liquid and gas with their behaviours.

Guided practice — answer in your exercise book

1. What usually keeps a fixed shape?
2. Which item is a solid?
3. What takes the shape of its container?
4. Which of these flows when poured?
5. What fills an inflated balloon?
6. Which state spreads to fill available space?
7. Which changes shape to fit a cup but keeps its amount?
8. Which has no fixed shape and can spread freely?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 6: Separating Solid Mixtures

Diversity of Matter | Recognise a solid–solid mixture and ways to separate its parts.

LET’S LEARN

A mixture contains materials placed together without making them all the same substance. Gari and groundnuts make a solid–solid mixture. Large beans can be picked by hand from clean rice. A sieve can separate larger particles from smaller ones, such as small pebbles from clean dry sand. Some metals can be attracted by a magnet, but magnets do not attract all metals. Choose a method based on the materials and their sizes or properties.

Science words to remember

Mixture • Handpicking • Sieving • Safe separation



Try it for yourself

With an adult, mix large clean dry beans and rice on a tray. Separate by hand and record how many beans you collected.

Safety note: Never use sharp nails, broken glass or unknown powders. Wash hands after handling mixtures.

QUEST 6: Show what you know

Observe the four teaching steps

1. MIX: Pour clean large beans and rice together.

2. PICK: Handpick the large beans from the tray.

3. SIEVE: Show differently sized dry particles passing through a sieve.

4. COMPARE: Explain when handpicking or sieving is more useful.

Guided practice — answer in your exercise book

1. What is gari mixed with groundnuts called?
2. Which has two different solids together?
3. How can large beans be separated from rice?
4. What helps select visible large pieces?
5. What separates large and small dry particles?
6. When does a sieve work well?
7. What should learners avoid when separating solids?
8. Who should supervise separation activities?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 7: Wet and Dry Seasons

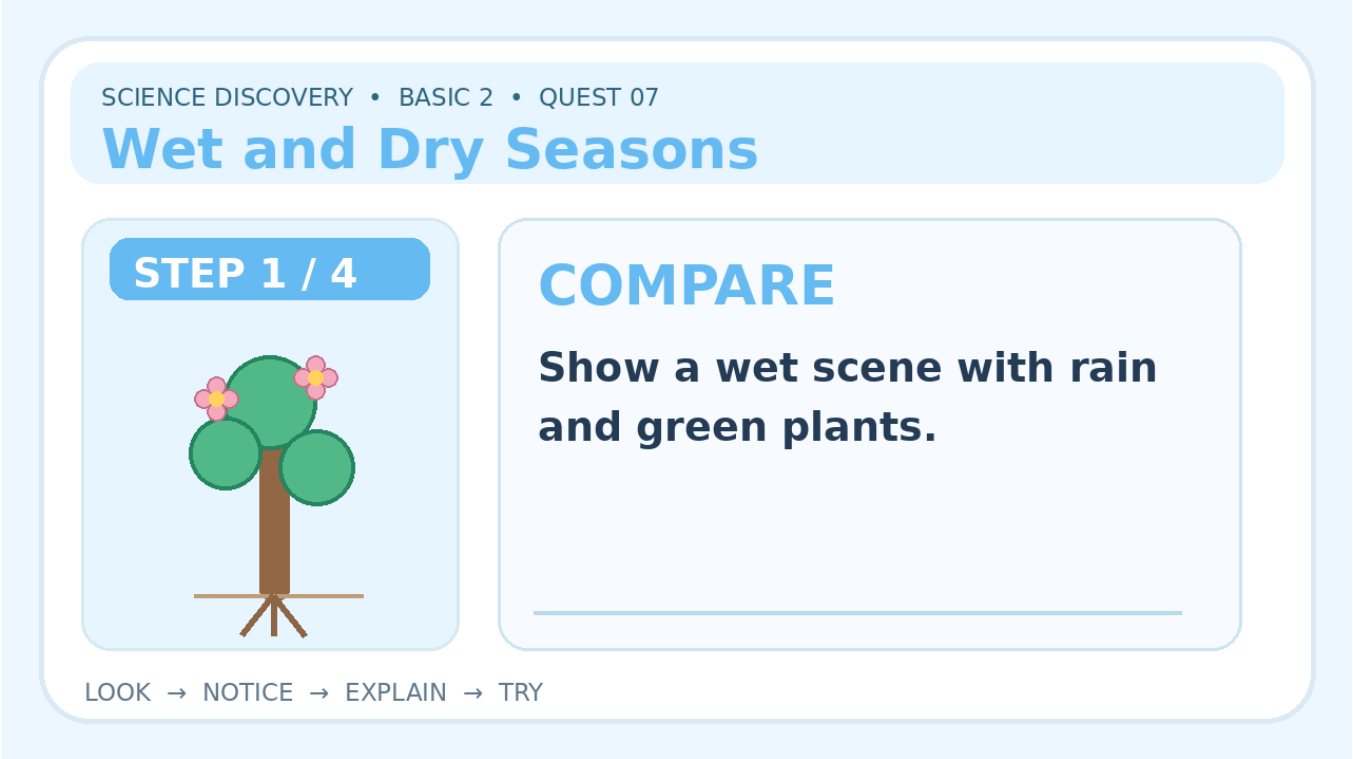
Cycles | Explain that wet and dry seasons recur and affect activities.

LET’S LEARN

Ghana has periods with more rain and periods with less rain. These are often called the wet and dry seasons. Seasons return as part of repeating natural patterns, but their timing and rainfall vary across Ghana. Farmers may plan planting when rain is reliable, while people may dry clothes outdoors on sunny days. During heavy rain, paths can become muddy. In dry months, collecting enough water may be difficult. Think about safe ways to prepare for either season.

Science words to remember

Wet season • Dry season • Repeating cycle • Preparation



Try it for yourself

Make two posters: a wet-season scene with rain and umbrellas, and a dry-season scene with dry clothes or stored water. Compare.

Safety note: Work with your teacher or a trusted adult.

QUEST 7: Show what you know

Observe the four teaching steps

1. COMPARE: Show a wet scene with rain and green plants.

2. COMPARE: Show a dry scene with sun and drier soil.

3. CYCLE: Connect wet and dry scenes with a repeating arrow.

4. PLAN: Choose an umbrella for rain and careful water use for dry days.

Guided practice — answer in your exercise book

1. Which season often brings more rain?
2. Why might a school path be muddy?
3. Which season often has less rain?
4. When may clothes dry faster outdoors?
5. What does it mean if seasons recur?
6. Which event is a repeating pattern?
7. What can help during a dry period?
8. What is safer during heavy rain?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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QUEST 8: Sources of Light

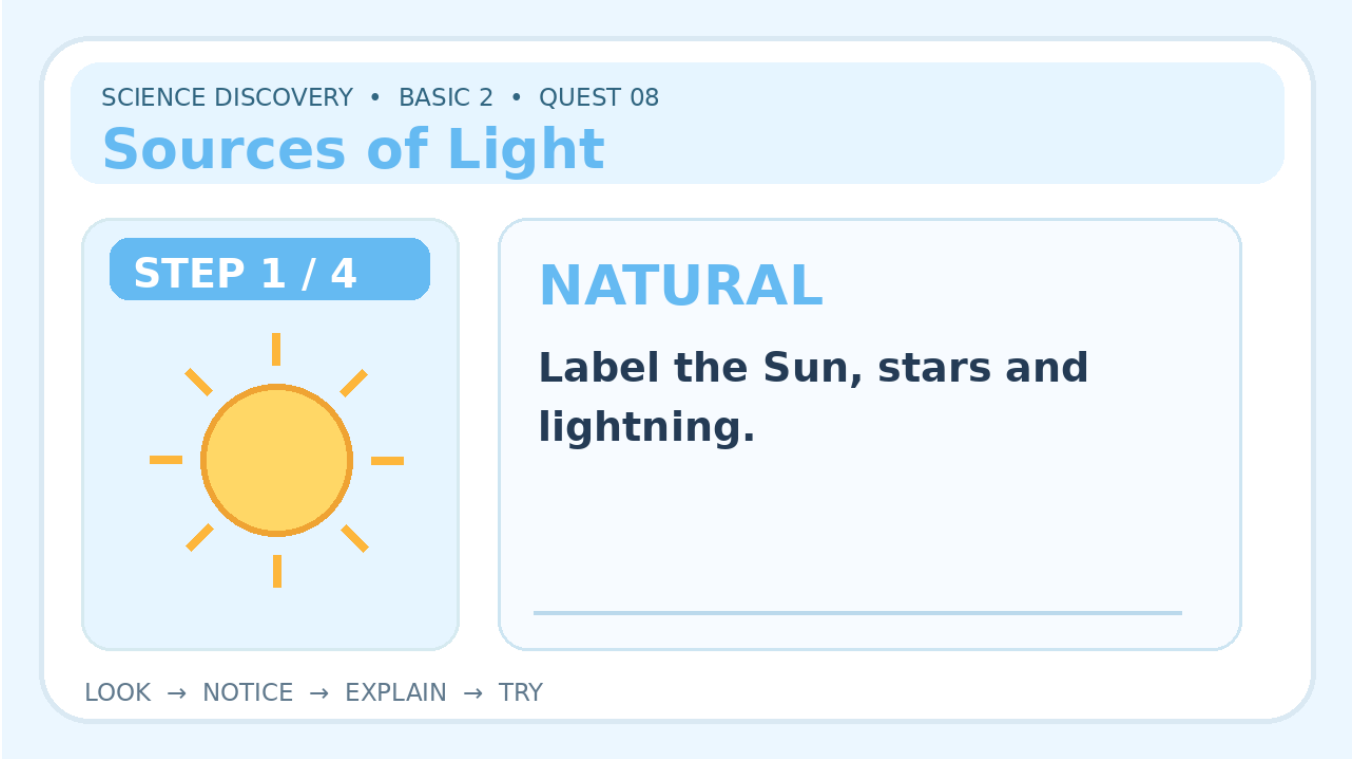
Cycles | Identify natural and human-made sources of light.

LET’S LEARN

The Sun, stars and lightning are natural sources of light. A light bulb, torch and lantern are human-made sources. Light allows us to see objects more clearly. A source makes or emits light; a mirror reflects light but does not make its own. Sunlight is important during the day, while a torch can help us see a dark room safely. Never look directly at the Sun or shine a strong light into someone’s eyes.

Science words to remember

Natural light • Artificial light • Reflectors • Light safety



Try it for yourself

Sort drawings of the Sun, stars, lamp, torch, mirror and firefly into light sources and objects that reflect light.

Safety note: Do not look directly at the Sun, use fire, or shine torches in eyes.

QUEST 8: Show what you know

Observe the four teaching steps

1. NATURAL: Label the Sun, stars and lightning.

2. ARTIFICIAL: Switch a drawing of a torch from off to on.

3. REFLECT: Show light rays bouncing from a mirror.

4. SORT: Place light sources into natural and artificial groups.

Guided practice — answer in your exercise book

1. Which is a natural source of light?
2. Which natural event can make a bright flash?
3. Which is a human-made light source?
4. What can you turn on in a dark room?
5. Does a mirror create light of its own?
6. Why can we see a mirror shining?
7. What should you never stare at directly?
8. What should you avoid doing with a torch?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 9: Choosing Safe Drinking Water

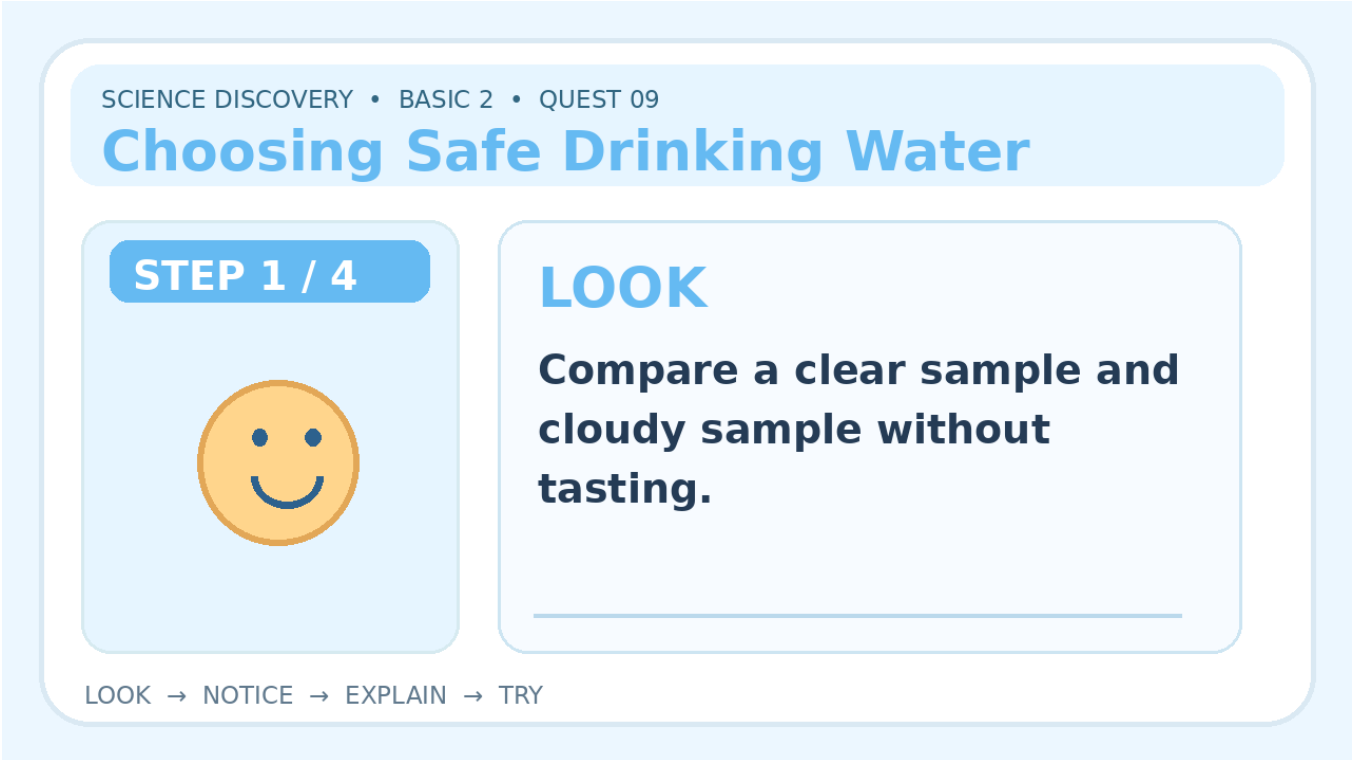
Cycles | Recognise observable qualities of drinking water and explain how to make safer choices.

LET’S LEARN

Water for drinking should be safely treated and stored. Good drinking water is usually clear, without visible particles or unusual colour or smell. However, even clear water can carry germs that we cannot see. We cannot decide water is safe by looking alone. Drink from a trusted treated supply. When unsure, ask an adult to use an approved treatment method such as proper boiling and safe storage; simple cloth filtering alone cannot reliably remove germs.

Science words to remember

Appearance • Invisible germs • Treatment • Safe storage



Try it for yourself

Observe only teacher-provided sealed water pictures or containers. Compare colour and visible particles. Do not taste any test sample.

Safety note: Never taste test samples. Boiling water and handling hot containers are adult-only tasks.

QUEST 9: Show what you know

Observe the four teaching steps

1. LOOK: Compare a clear sample and cloudy sample without tasting.

2. ZOOM: Show germs too small to be seen by the eye.

3. TREAT: Show a trusted adult making water safer.

4. STORE: Keep treated water in a clean covered vessel.

Guided practice — answer in your exercise book

1. Which water appears clear?
2. What might be seen in dirty water?
3. Can clear-looking water still contain germs?
4. Can you prove water is safe by sight alone?
5. What helps make questionable water safer?
6. Is filtering through cloth alone enough to kill germs?
7. Where should safe drinking water be stored?
8. Should learners taste classroom water samples?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 10: How We Use Air

Cycles | Describe everyday uses of air, including breathing, inflating and wind.

LET’S LEARN

Air is all around us. We cannot normally see it, but we can notice what it does. People and animals breathe air. Moving air is called wind; it can fly a kite, turn a pinwheel and cool a wet surface. Pumping air into a tyre gives it firmness. Air can also fill a balloon. In class we can observe air without blowing into or sharing objects unsafely.

Science words to remember

Breathing • Inflating • Wind • Observing



Try it for yourself

With an adult, use a hand fan to move a paper pinwheel. Observe what happens as you fan gently and then stop.

Safety note: Use clean personal materials. Avoid inflating balloons by mouth if there is choking risk; adults supervise.

QUEST 10: Show what you know

Observe the four teaching steps

1. BREATHE: Show a child breathing normally.

2. INFLATE: Fill a cartoon balloon and bicycle tyre with air.

3. WIND: Move a paper pinwheel with airflow arrows.

4. OBSERVE: Compare still air and moving air.

Guided practice — answer in your exercise book

1. Why do people need air?
2. What do lungs take in when we breathe?
3. What fills an inflated balloon?
4. Why is air pumped into a bicycle tyre?
5. What is moving air called?
6. What helps a kite fly on a windy day?
7. How can we notice invisible air?
8. What happens to a pinwheel when fanning stops?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 11: What Our Body Parts Do

Systems | Connect eyes, ears, nose, mouth, hands, arms and legs to their functions.

LET’S LEARN

We use body parts together to do everyday activities. Eyes help us see. Ears help us hear. The nose helps us smell and breathe. The mouth helps us speak and eat. Hands and arms help us hold, write, lift and reach. Legs help us stand, walk and run. People have different abilities, and assistive devices may help them do these tasks. We must respect all bodies.

Science words to remember

Eyes and ears • Nose and mouth • Arms and hands • Legs



Try it for yourself

Play a safe function-matching game: point to a body part and name a job it helps us do. Draw and label a person.

Safety note: Work with your teacher or a trusted adult.

QUEST 11: Show what you know

Observe the four teaching steps

1. SEE AND HEAR: Point to eyes and ears on a friendly diagram.

2. SMELL AND SPEAK: Highlight nose and mouth.

3. HOLD: Show hands grasping a light pencil.

4. MOVE: Show legs walking safely.

Guided practice — answer in your exercise book

1. Which part helps us see?
2. Which part helps us hear sounds?
3. Which part helps us smell?
4. Which part helps us speak words?
5. What helps us hold a pencil?
6. Which parts help us lift light objects?
7. Which parts support walking?
8. What helps us jump with safe balance?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 12: Sunlight and Warmth

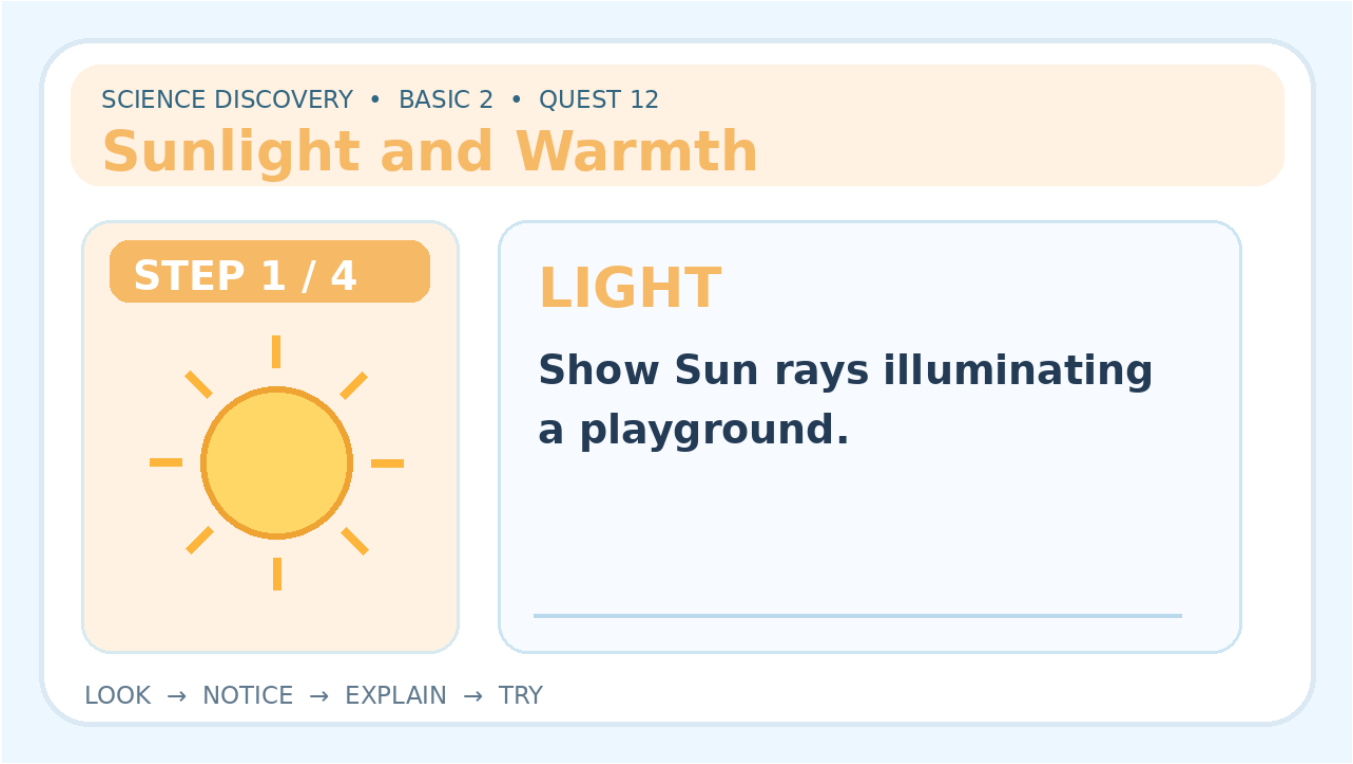
Systems | Identify the Sun as the main natural source of light and warmth for Earth.

LET’S LEARN

The Sun is Earth’s main natural source of daytime light and warmth. It helps green plants make food and warms land and water. Shade can protect us from too much direct sunlight. At night, we do not see the Sun because our part of Earth faces away from it. The Moon can appear bright by reflecting sunlight, but it does not create sunlight of its own.

Science words to remember

Light • Warmth • Shade • Reflection



Try it for yourself

Compare pictures of a sunny playground and a shaded space. Record where a person might feel cooler.

Safety note: Do not look directly at the Sun; avoid prolonged heat exposure.

QUEST 12: Show what you know

Observe the four teaching steps

1. LIGHT: Show Sun rays illuminating a playground.

2. WARM: Show sunlight reaching the ground.

3. SHADE: Compare under-tree shade and direct sun.

4. REFLECT: Moon reflects Sun’s light at night.

Guided practice — answer in your exercise book

1. What is our main daytime natural light source?
2. What helps us see clearly outdoors by day?
3. What warms the ground on a sunny day?
4. Why might a shaded place feel cooler?
5. What can protect us from very strong sunlight?
6. Where might a child rest at midday?
7. Why does the Moon look bright?
8. Does the Moon create its own sunlight?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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QUEST 13: Energy in Everyday Life

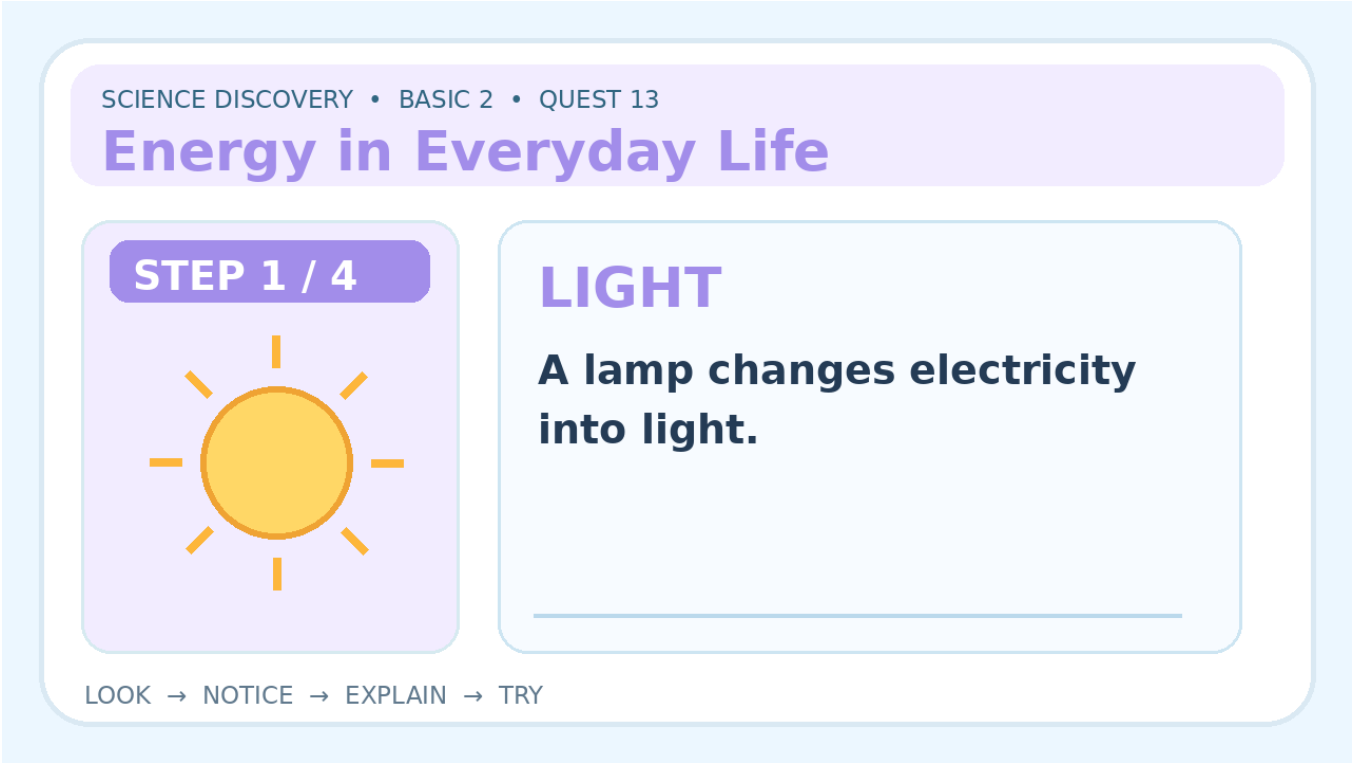
Forces and Energy | Identify everyday uses of energy in homes, school and the community.

LET’S LEARN

Energy helps things happen. People use energy from food to play, walk and work. Sunlight gives light and warmth. Electricity can power a fan, lamp or radio. A battery can make a torch shine. We use energy in many forms, such as light, heat, movement and sound. Looking at what a device does helps us recognise the type of energy we see or hear.

Science words to remember

Light • Heat • Movement • Sound



Try it for yourself

Draw a sun, lamp, radio and moving bicycle. Say what each one does with energy.

Safety note: Work with your teacher or a trusted adult.

QUEST 13: Show what you know

Observe the four teaching steps

1. LIGHT: A lamp changes electricity into light.

2. HEAT: A warm meal shows heat being used in cooking.

3. MOVE: Fan blades spin and a child pedals a bicycle.

4. SOUND: A drum or radio sends out sound.

Guided practice — answer in your exercise book

1. What form of energy helps us see a lamp?
2. What does the Sun give in daytime?
3. What form of energy warms food during cooking?
4. What can warmth do to a cold object?
5. What do legs use energy to produce?
6. What can a fan use electricity to produce?
7. What comes from a ringing school bell?
8. What form of energy helps us hear a drum?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

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QUEST 14: Gaining and Losing Heat

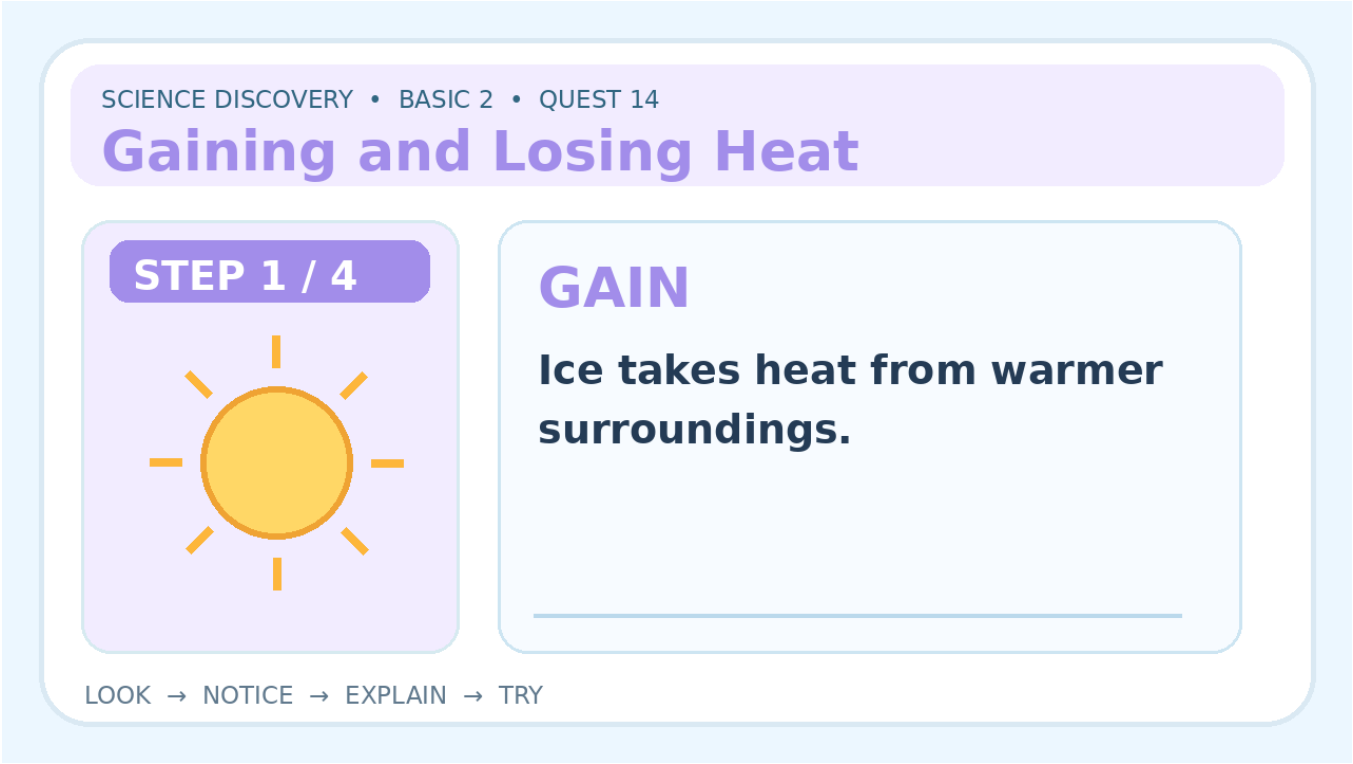
Forces and Energy | Explain how objects become warmer by gaining heat or cooler by losing it.

LET’S LEARN

Heat is transferred from a warmer object or place to a cooler one. A cup of tea can cool as it loses heat to the surroundings. An ice cube warms and melts as it gains heat. A cold spoon placed in warm (not hot) water becomes warmer. The direction of heat movement helps us explain everyday changes. Temperatures do not change because a material is happy or sad!

Science words to remember

Gaining heat • Losing heat • Warmer and cooler • Safe observation



Try it for yourself

With an adult, place a safe sealed cool bottle in a warm room. Observe it over time without handling boiling water.

Safety note: Do not handle hot water, irons, stove burners or other heated objects.

QUEST 14: Show what you know

Observe the four teaching steps

1. GAIN: Ice takes heat from warmer surroundings.

2. MELT: The ice gradually turns into liquid water.

3. LOSE: A warm drink loses heat and cools.

4. COMPARE: Arrows show heat moving from warm towards cool.

Guided practice — answer in your exercise book

1. What happens when an object gains heat?
2. Why may ice melt in a warm room?
3. What happens as hot tea loses heat?
4. What happens to a warm cup left on a cool table?
5. Where does heat usually flow naturally?
6. When a cool spoon warms in warm water, it has ...?
7. Which is safe to observe?
8. Who should handle hot cooking water?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 15: Electrical Safety First

Forces and Energy | Recognise safe and unsafe ways of using electricity.

LET’S LEARN

Electricity powers many devices, but electric current can harm people if used carelessly. Never touch broken wires or sockets, or use electric items with wet hands. Children should not repair plugs or connect appliances to wall sockets without an adult. Keep water away from electrical equipment. Tell a trusted adult if you notice sparks, burning smells or exposed wires. Safe choices prevent serious accidents.

Science words to remember

Wet hands • Broken wires • Adult help • Water safety



Try it for yourself

Sort teacher-drawn cards into SAFE and UNSAFE. Practise telling an adult when you notice an unsafe plug in a picture.

Safety note: No hands-on work with mains electricity, switches, sockets or exposed wiring.

QUEST 15: Show what you know

Observe the four teaching steps

1. SAFE: A child keeps dry hands near an approved switch.

2. DANGER: Highlight exposed wires with a stop symbol.

3. DANGER: Show water beside plugged-in devices.

4. ACT: Child moves away and informs a responsible adult.

Guided practice — answer in your exercise book

1. Should you use an electric switch with wet hands?
2. What is safer before touching approved equipment?
3. What should you do if a wire is exposed?
4. Is a broken wall socket safe to touch?
5. Who should repair an electrical plug?
6. What should a child do when an appliance sparks?
7. Why keep water away from electric devices?
8. Which is unsafe?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 16: Batteries Power Devices

Forces and Energy | Describe what batteries do in simple electronic devices.

LET’S LEARN

A battery stores energy and supplies electrical energy to a suitable device. A battery-powered torch can shine when its battery is fitted correctly. Some toy cars move using batteries. If the battery is removed or used up, the device may stop. Batteries have positive and negative ends that must be placed correctly, according to a safe product’s instructions. Do not open batteries or put them in your mouth; leaking batteries are dangerous.

Science words to remember

Power source • Working torch • Battery removal • Safe use



Try it for yourself

With an adult, inspect a battery-powered torch without removing batteries yourself. Draw the torch and label its power source.

Safety note: Do not open batteries or attempt circuits without direct qualified adult supervision.

QUEST 16: Show what you know

Observe the four teaching steps

1. SHOW: A torch and toy car are placed on a table.

2. POWER: Label the battery compartment safely.

3. COMPARE: Working torch lights; removed battery makes it go off.

4. SAFETY: Display no-opening and no-chewing battery warnings.

Guided practice — answer in your exercise book

1. What gives a battery-powered torch electrical energy?
2. Which can power a small toy car?
3. What usually happens when a torch has a working battery and is switched on?
4. What might stop a working toy when its battery is taken out?
5. Why might an old battery fail to power a torch?
6. What should a child do when a toy stops working?
7. What should we never do to a battery?
8. What should an adult do with leaking batteries?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 17: How Forces Change Movement

Forces and Energy | Investigate forces that start, stop, change direction or shape objects.

LET’S LEARN

A force is a push or pull. A kick can start a football moving. Catching the ball can stop it. A push from another direction can turn a rolling ball. Squeezing a sponge changes its shape. Pulling a toy on a string can bring it closer. Different forces can change how an object moves or looks. Use soft safe objects for experiments.

Science words to remember

Start movement • Stop movement • Change direction • Change shape



Try it for yourself

Roll a soft ball. With an adult, stop it using two hands, then gently change its direction. Write down what each force did.

Safety note: Use a soft ball and safe open space; never kick towards faces or traffic.

QUEST 17: Show what you know

Observe the four teaching steps

1. START: Show a ball at rest, then a foot gently kicking it.

2. STOP: Two hands stop the rolling ball.

3. TURN: A side push changes the ball direction.

4. SHAPE: A soft sponge changes shape under gentle squeeze.

Guided practice — answer in your exercise book

1. What can start a stationary ball moving?
2. Which action pushes a chair?
3. What can stop a rolling soft ball?
4. What happens when a moving toy is held safely?
5. What can turn a moving ball to the side?
6. What can change the direction of a rolling toy?
7. What can change the shape of a sponge?
8. Which object can be squashed safely?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 18: Simple Machines at Work

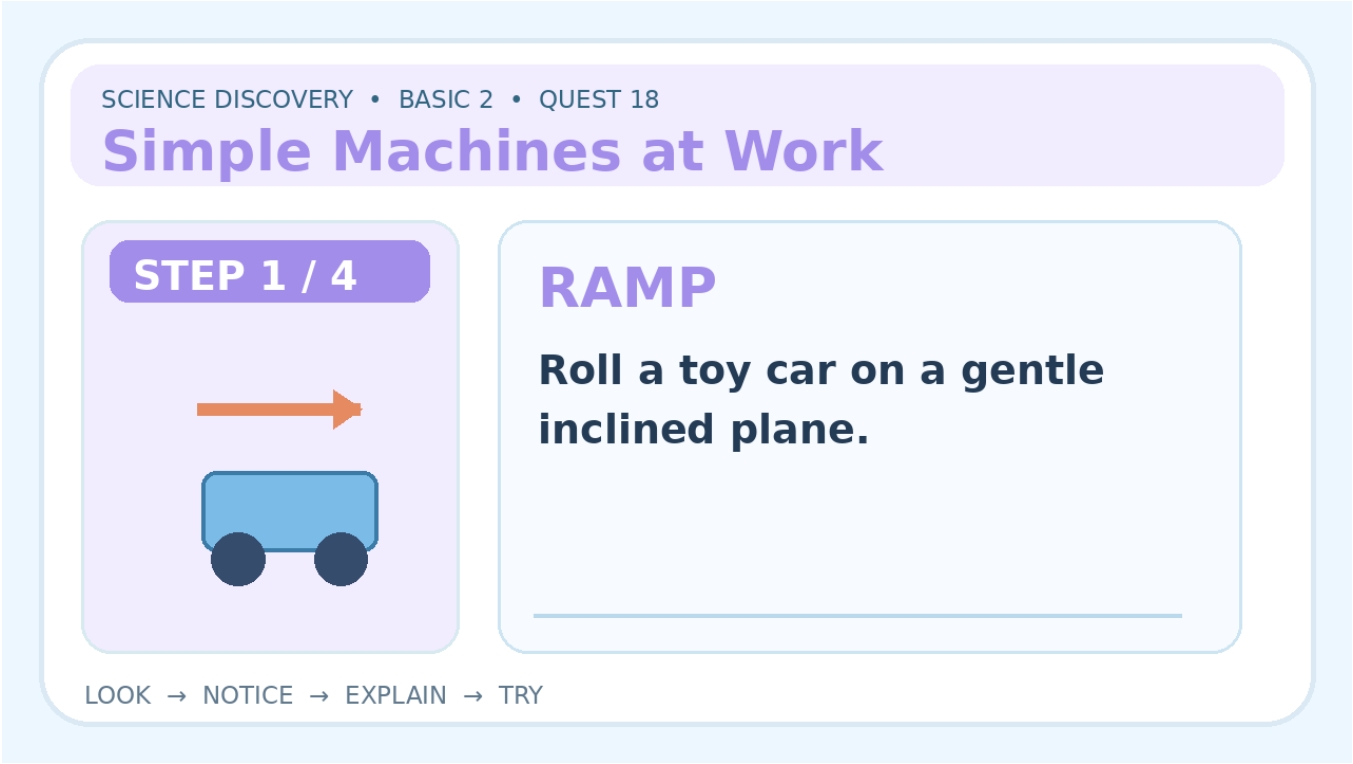
Forces and Energy | Recognise simple machines and how they make tasks easier.

LET’S LEARN

Simple machines help people do work. A ramp, called an inclined plane, can help move a heavy object up to a platform with less effort over a longer distance. A lever helps lift or open things; a bottle opener is one example. Wheels on a trolley make it easier to move goods. A pulley can help lift a bucket, when correctly used by an adult. We choose the machine that suits a job.

Science words to remember

Inclined plane • Lever • Wheel • Pulley



Try it for yourself

Build a small cardboard ramp and gently roll a toy car up it. Draw the ramp and compare the path with lifting the toy straight up.

Safety note: Never lift heavy loads or use tools with sharp edges without trained adult supervision.

QUEST 18: Show what you know

Observe the four teaching steps

1. RAMP: Roll a toy car on a gentle inclined plane.

2. LEVER: Show a bottle opener lifting a lid (adult demonstration).

3. WHEEL: Show a trolley carrying boxes on wheels.

4. PULLEY: Show a rope and wheel lifting a small bucket safely.

Guided practice — answer in your exercise book

1. What is another name for a ramp?
2. Which helps push a box gradually up to a platform?
3. Which simple machine helps pry open some lids?
4. What is an example of a lever?
5. What makes a trolley roll easily?
6. Which machine helps move goods along the floor?
7. What simple machine may help lift a bucket?
8. What should you use to raise a flag with a rope on a suitable pole?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 19: Keep Your Body Clean

Humans and the Environment | Explain how good hygiene keeps our body clean and healthy.

LET’S LEARN

Bathing with clean water and appropriate soap removes sweat and dirt. Washing hands with soap helps remove germs that can make us ill. Brush teeth gently, wear clean clothes and keep fingernails short and clean. Clean towels and personal bathing items should not be shared carelessly. We can develop a daily routine to care for our bodies.

Science words to remember

Bathing • Handwashing • Tooth care • Clean clothing



Try it for yourself

Make a morning-and-evening hygiene timetable. Tick bathing, handwashing, toothbrushing and clean clothing activities with help at home.

Safety note: Work with your teacher or a trusted adult.

QUEST 19: Show what you know

Observe the four teaching steps

1. BATH: Show safe washing with soap and clean water.

2. HANDS: Demonstrate soap rubbing and rinsing.

3. TEETH: Show gentle toothbrushing.

4. DRESS: Choose clean clothes and personal towels.

Guided practice — answer in your exercise book

1. What helps remove sweat from skin?
2. Why should we bathe regularly?
3. When should we wash hands with soap?
4. What helps remove germs from hands?
5. What helps care for teeth daily?
6. What is a good time to brush teeth?
7. What clothes help keep a body clean?
8. Why change sweaty clothes when possible?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 20: Caring for a Clean Classroom

Humans and the Environment | Describe and practise safe ways to keep school surroundings clean.

LET’S LEARN

A tidy classroom makes it easier to learn and move safely. We can sweep floors, arrange learning materials neatly and put safe litter into bins. Dirty standing water around school may provide places for mosquitoes to breed. Keeping drains clear and seeking adult help for risky waste can protect everyone. Good ventilation helps fresh air move indoors. Keeping school clean is a shared responsibility.

Science words to remember

Sweeping • Bins • Mosquito prevention • Fresh air



Try it for yourself

With your teacher, sort picture cards showing safe sweeping, bin use, open windows and unsafe dumping. Draw a clean classroom.

Safety note: Do not touch broken glass, human waste, sharp items or unknown liquids; tell an adult.

QUEST 20: Show what you know

Observe the four teaching steps

1. BEFORE: Litter on a floor and near a drain.

2. CLEAN: Sweep safe paper litter into a bin.

3. CHECK: Remove standing water with adult help.

4. AFTER: Neat room with fresh air through windows.

Guided practice — answer in your exercise book

1. What helps remove dust from a classroom floor?
2. What should we do with classroom litter?
3. Where should a paper wrapper go?
4. What must we avoid leaving around school?
5. Why clear places where mosquitoes breed?
6. What helps fresh air enter a safe classroom?
7. Why is ventilation useful?
8. Who helps keep a classroom tidy?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 21: Ringworm and Healthy Skin

Humans and the Environment | Identify ringworm as a fungal infection and describe prevention.

LET’S LEARN

Ringworm is a skin infection caused by fungi, not by a worm. It may cause a round, itchy, scaly patch, but other conditions can look similar. Ringworm can spread through close contact or sharing contaminated items. Avoid sharing towels or combs, keep skin clean and dry, and tell a trusted adult if a rash appears. A health worker can decide if treatment is needed. Eczema is different and is not contagious.

Science words to remember

Cause • Appearance • Prevention • Asking for help



Try it for yourself

Sort pictures of healthy hygiene habits: personal towels, dry clothes and adult care. Do not diagnose classmates.

Safety note: No student should diagnose or touch another learner’s rash or share towels for demonstrations.

QUEST 21: Show what you know

Observe the four teaching steps

1. CAUSE: Show a labelled fungus icon, not a worm.

2. LOOK: A simple drawing of a possible round skin rash.

3. PREVENT: Use personal towels and clean clothing.

4. HELP: Tell a trusted adult and see a healthcare worker.

Guided practice — answer in your exercise book

1. What causes ringworm?
2. Is ringworm caused by a real ring-shaped worm?
3. What may ringworm sometimes look like?
4. Can every circular rash be identified as ringworm by a child?
5. What helps reduce spread?
6. What should children keep clean and dry?
7. What should a child do about a spreading rash?
8. Who can help diagnose a skin infection?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 22: Protecting Against Water-borne Diseases

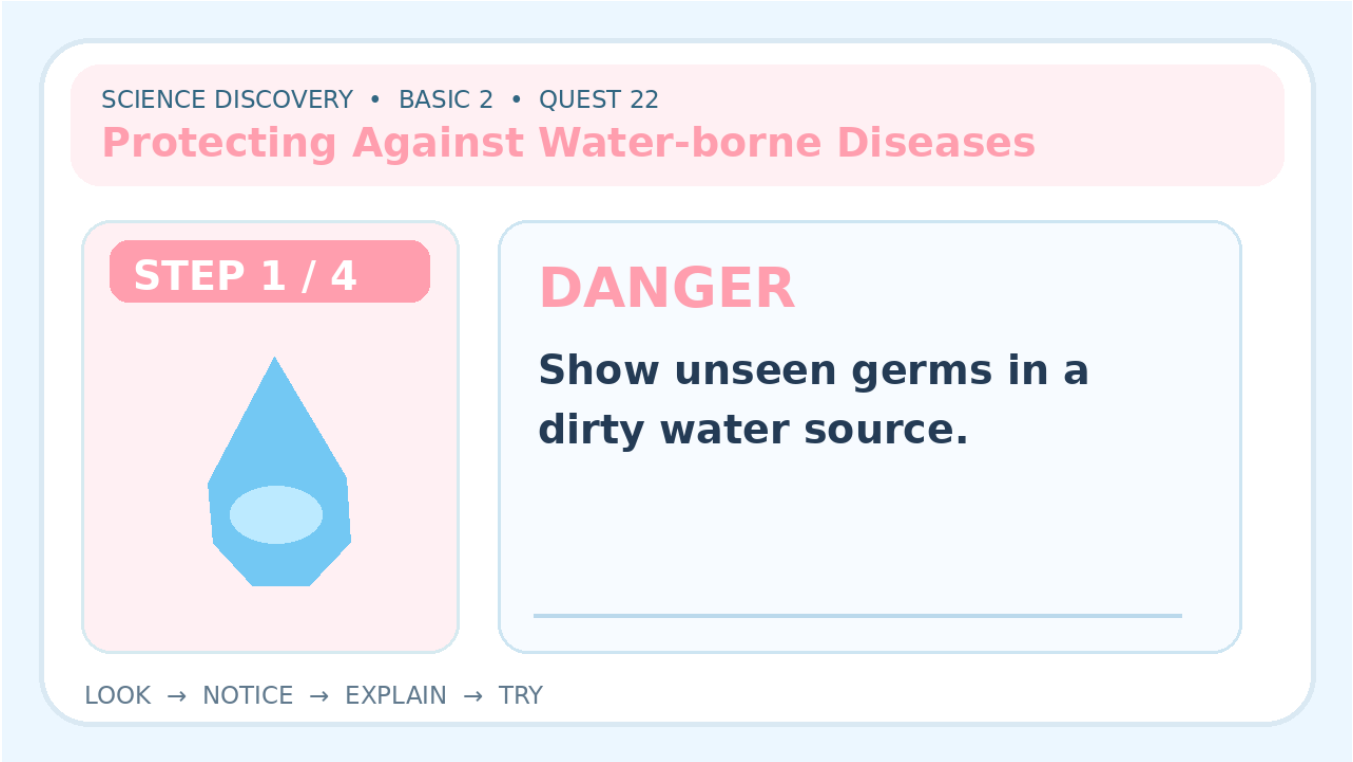
Humans and the Environment | Recognise common water-borne diseases and prevention methods.

LET’S LEARN

Drinking water contaminated with harmful germs can cause diseases such as cholera, typhoid fever and some diarrhoeal illnesses. Water may look clear and still contain germs. Safe drinking water, proper sanitation, good handwashing and safe food handling lower risks. An adult can make unsafe water safer using appropriate treatment, such as proper boiling. A child who has severe diarrhoea or signs of dehydration needs prompt adult and medical attention.

Science words to remember

Diseases • Contaminated water • Prevention • Help



Try it for yourself

Make a safety poster showing clean covered drinking water, handwashing and proper toilet use. Do not taste test water.

Safety note: Never taste classroom water samples or boil water without adult help.

QUEST 22: Show what you know

Observe the four teaching steps

1. DANGER: Show unseen germs in a dirty water source.

2. DISEASE: Label cholera, typhoid and diarrhoeal illness.

3. PREVENT: Use safe treated water and handwashing.

4. HELP: Tell adult early about severe illness or dehydration.

Guided practice — answer in your exercise book

1. Which illness can spread through contaminated water?
2. Which is another water-borne disease?
3. What may make drinking water unsafe?
4. Can clear-looking water contain harmful germs?
5. How can families lower the risk of water-borne illnesses?
6. What hygiene habit protects against germs?
7. What should a child do if very unwell with diarrhoea?
8. Who should handle boiling treatment of water?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 23: Technology in Our Community

Humans and the Environment | Recognise modern devices and explain how they help people.

LET’S LEARN

Technology includes tools and devices designed to solve problems. A mobile phone helps people communicate over distances. A computer can store and organise information. A sewing machine helps make clothes. Hospital tools help health workers care for patients. Vehicles help move people and goods. People used other methods long ago, such as letters and walking long distances; compare how technology changes activities, while remembering it must be used safely.

Science words to remember

Communication • Work • Healthcare • Travel



Try it for yourself

Ask an adult about one device used in the past and a newer alternative. Draw them and name one advantage and one safe-use rule.

Safety note: Work with your teacher or a trusted adult.

QUEST 23: Show what you know

Observe the four teaching steps

1. PAST: Show a handwritten letter carried by hand.

2. NOW: Show people contacting family with a phone.

3. WORK: Show sewing machine and computer at work.

4. TRAVEL: Show bus or truck helping the community.

Guided practice — answer in your exercise book

1. What device helps people talk over long distances?
2. What older way could people send distant messages?
3. What helps a tailor stitch cloth?
4. What helps people store information?
5. Where are some medical technologies used?
6. Who may use equipment to examine patients?
7. What helps move many passengers along roads?
8. What helps move goods to market?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 24: Preparing Food Safely

Humans and the Environment | Identify basic methods used to process food for eating.

LET’S LEARN

Foods can be prepared in different ways. Washing helps remove some dirt from vegetables, while proper cooking reduces many germs. Boiling cooks foods in hot water, roasting uses dry heat, frying cooks in hot oil and baking uses heated oven air. Some foods can be eaten raw after safe preparation. Children can identify these methods in pictures but should not handle sharp knives, open flames, boiling water or hot oil.

Science words to remember

Washing • Boiling • Roasting • Frying and baking



Try it for yourself

Create picture cards for washing, boiling, roasting, frying and baking. Group everyday Ghanaian foods under suitable preparation methods.

Safety note: Cooking flames, knives, ovens, boiling water and hot oil are adult-only demonstrations.

QUEST 24: Show what you know

Observe the four teaching steps

1. WASH: Show vegetables rinsed with safe water.

2. BOIL: An adult cooks yam in hot water.

3. ROAST: An adult roasts maize over controlled heat.

4. FRY OR BAKE: Label hot-oil frying and oven baking as adult cooking methods.

Guided practice — answer in your exercise book

1. What is a first safe step for many vegetables?
2. What does washing help remove from food?
3. How is yam often cooked in water?
4. What cooks food in hot water?
5. What method uses dry heat to brown maize?
6. What is food cooked over dry heat called?
7. Which method uses hot oil?
8. Which method often uses a heated oven?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

QUEST 25: Protect Our Environment

Humans and the Environment | Identify harmful human activities and explain safer environmental choices.

LET’S LEARN

People can harm the environment by burning rubbish, cutting trees without replanting, dumping litter into rivers and polluting the air. Smoke may harm breathing, and contaminated water may make people sick. Fuels such as petrol release gases that contribute to global warming when burned. Communities can reduce waste, protect water, plant and care for trees, and choose cleaner habits. Small safe actions help our school and Ghana.

Science words to remember

Smoke pollution • Water pollution • Warming • Solutions



Try it for yourself

Draw two community scenes, one with unsafe dumping and smoke and one with bins, clean water and cared-for trees. Explain three differences.

Safety note: Do not approach fires or polluted water; environmental cleanups need adult guidance.

QUEST 25: Show what you know

Observe the four teaching steps

1. POLLUTE AIR: Show open rubbish burning with smoke.

2. POLLUTE WATER: Show litter entering a stream.

3. WARM: Explain fuel burning releases warming gases.

4. PROTECT: Show bins, cleaner streams and planted trees.

Guided practice — answer in your exercise book

1. What may happen when rubbish burns in open air?
2. Why avoid breathing smoky air?
3. What may pollute a river?
4. Why must water bodies be protected?
5. What can burning petrol add to the atmosphere?
6. What global effect can some greenhouse gases increase?
7. What helps keep a community cleaner?
8. What supports healthy green surroundings?

Practical reflection

Draw your observation or make a small labelled diagram. Explain your result to a partner or trusted adult.

Remember: the online course contains 10 play questions and 25 challenge questions for this lesson.

Teacher and parent answer key

Quest 1: Parts of a Green Plant

1. Roots • 2. Roots • 3. Stem • 4. Stem • 5. Sunlight • 6. Leaf • 7. Flower • 8. Above the soil

Quest 2: Animal Body Parts

1. On its head • 2. Head • 3. Trunk • 4. Trunk • 5. Walk and run • 6. Two • 7. No • 8. Chicken

Quest 3: Living or Non-living?

1. A growing seedling • 2. Mango plant • 3. A desk • 4. Toy wheel • 5. It grows and needs food • 6. It does not carry out life processes • 7. Living group • 8. Non-living group

Quest 4: Explore Material Properties

1. Cotton wool • 2. Stone • 3. Sandpaper • 4. Polished tile • 5. Clean transparent plastic • 6. Wooden board • 7. Bend • 8. Soft rubber band

Quest 5: Solids, Liquids and Gases

1. Stone • 2. Wooden block • 3. Water • 4. Clean water • 5. Air • 6. Gas • 7. Liquid water • 8. Air

Quest 6: Separating Solid Mixtures

1. A mixture • 2. Beans and rice • 3. Handpicking • 4. Clean fingers or tongs • 5. Sieve • 6. When particles differ in size • 7. Sharp nails • 8. Teacher or adult

Quest 7: Wet and Dry Seasons

1. Wet season • 2. Rain has soaked the ground • 3. Dry season • 4. On a sunny dry day • 5. They come again • 6. Wet and dry seasons • 7. Use water carefully • 8. Walk carefully and stay sheltered

Quest 8: Sources of Light

1. Sun • 2. Lightning • 3. Electric bulb • 4. Torch • 5. No, it reflects light • 6. It reflects light • 7. Sun • 8. Shining it into eyes

Quest 9: Choosing Safe Drinking Water

1. Water without visible particles • 2. Suspended particles • 3. Yes • 4. No • 5. Approved treatment by an adult • 6. No • 7. In a clean covered container • 8. No

Quest 10: How We Use Air

1. To breathe • 2. Air • 3. Air • 4. To make it firm • 5. Wind • 6. Moving air • 7. See a pinwheel turn • 8. It may slow or stop

Quest 11: What Our Body Parts Do

1. Eyes • 2. Ears • 3. Nose • 4. Mouth • 5. Hands • 6. Arms and hands • 7. Legs • 8. Legs

Quest 12: Sunlight and Warmth

1. Sun • 2. Sunlight • 3. Sun • 4. It receives less direct sunlight • 5. Shade • 6. Under safe shade • 7. It reflects sunlight • 8. No

Quest 13: Energy in Everyday Life

1. Light • 2. Light • 3. Heat • 4. Make it warmer • 5. Movement • 6. Movement of its blades • 7. Sound • 8. Sound

Quest 14: Gaining and Losing Heat

1. It may become warmer • 2. It gains heat • 3. It becomes cooler • 4. It cools gradually • 5. From warmer to cooler • 6. Gained heat • 7. A sealed bottle warming naturally • 8. A responsible adult

Quest 15: Electrical Safety First

1. No • 2. Dry hands • 3. Tell an adult and keep away • 4. No • 5. Qualified adult • 6. Move away and tell an adult • 7. Water increases electric-shock risk • 8. Using an appliance near spilled water

Quest 16: Batteries Power Devices

1. Battery • 2. Suitable battery • 3. It shines • 4. Its motor or lights • 5. Its stored energy is used up • 6. Ask an adult to check the battery • 7. Open or chew it • 8. Handle and dispose of safely

Quest 17: How Forces Change Movement

1. A gentle kick • 2. Pressing it forward safely • 3. Catching it safely • 4. It stops • 5. Push from another direction • 6. Sideways push • 7. Squeezing • 8. Soft sponge

Quest 18: Simple Machines at Work

1. Inclined plane • 2. Ramp • 3. Lever • 4. Bottle opener • 5. Wheels • 6. Wheeled cart • 7. Pulley • 8. Pulley

Quest 19: Keep Your Body Clean

1. Bathing with soap and water • 2. To remove dirt and sweat • 3. Before eating • 4. Soap and water • 5. Gentle brushing • 6. Before bed • 7. Clean clothes • 8. To stay clean and comfortable

Quest 20: Caring for a Clean Classroom

1. Sweeping safely • 2. Put it in a bin • 3. Suitable bin • 4. Standing dirty water • 5. To reduce mosquitoes • 6. Open suitable windows • 7. It allows fresh air movement • 8. Everyone working safely

Quest 21: Ringworm and Healthy Skin

1. Fungus • 2. No • 3. Round itchy scaly patch • 4. No • 5. Do not share personal towels • 6. Skin and personal items • 7. Tell an adult and seek health advice • 8. Health worker

Quest 22: Protecting Against Water-borne Diseases

1. Cholera • 2. Typhoid fever • 3. Harmful germs • 4. Yes • 5. Drink safely treated water • 6. Washing hands with soap • 7. Tell an adult for prompt medical help • 8. Responsible adult

Quest 23: Technology in Our Community

1. Mobile phone • 2. Letters • 3. Sewing machine • 4. Computer • 5. Hospital • 6. Health worker • 7. Bus • 8. Truck

Quest 24: Preparing Food Safely

1. Wash using safe water • 2. Some dirt • 3. Boiling • 4. Boiling • 5. Roasting • 6. Roasted food • 7. Frying • 8. Baking

Quest 25: Protect Our Environment

1. Smoke pollutes air • 2. It can harm our lungs • 3. Dumping rubbish into it • 4. Dirty water can harm people and animals • 5. Greenhouse gases • 6. Warming of Earth • 7. Use bins and protect streams • 8. Care for planted trees

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